

ECO387C - Fall 2024

Problem Set 11

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You may work in groups, but please acknowledge any contributions from others and turn in separate assignments.

Question 1 - Solow model without technological change

Consider the capital accumulation equation in *continuous time*

$$\dot{k}(t) = sf(k(t)) - (n + \delta)k(t)$$

where $k(t)$ is the capital per effective labor (i.e. $k(t) \equiv \frac{K(t)}{A(t)L(t)}$) and $f(k(t))$ is the neoclassical production function in intensive form. Let $k(t)$ be the solution to this equation with the initial condition $k(0) = k_0 > 0$. Let $w(t)$ and $R(t)$ be the corresponding equilibrium factor prices at date t .

1. Show that $w(t)$ and $R(t)$ are monotone functions of time.
2. Let k^* be the steady-state capital-labor ratio. Sketch $k(t)$, $y(t)$, $c(t)$, $w(t)$, and $R(t)$ for a given $k_0 < k^*$. Repeat the same for a given $k_0 > k^*$.
3. Assume that the production function is Cobb-Douglas:

$$F(K, L, A) = AK^\alpha L^{1-\alpha}$$

Derive the corresponding intensive-form production function $f(k)$. Solve the capital accumulation equation given the initial condition $k(0) = k_0 > 0$. Compare your solution to the sketch in part (b).

(Hint 1: use the change of variables $x(t) = k(t)^{1-\alpha}$.)

(Hint 2: For a class of linear differential equation $\dot{x}(t) = a + bx(t)$, Guess a solution of the form $x = -\frac{a}{b} + c \exp(bt)$ where a, b, c are some constants)

Question 2 - Solow model with technological growth

Consider the Solow model with technological growth:

$$\dot{K}_t = sK_t^\alpha (A_t L_t)^{1-\alpha} - \delta K_t$$

$$\dot{A} = g_A \cdot A_t$$

$$\dot{L} = nL_t$$

Sketch the time paths for $w(t)$, $R(t)$, $K(t)$, $Y(t)$ (or it may be easier to consider the paths for $\log w(t)$, $\log K(t)$, $\log Y(t)$) for the following parameter change scenarios:

1. A permanent increase in s .
2. A permanent increase in n .
3. A permanent increase in g_A .